Enhancing Face-API.js Emotion Recognition Performance for Sri Lankan Cultural Demographic Faces and DeepFake Image Detection.

R26-IT-011

Project Proposal Report

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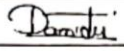
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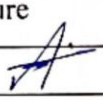
March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Emotion recognition systems have become foundational pillars in advancing Human-Computer Interaction (HCI), ranging from therapeutic mental health monitoring to high-security surveillance and intelligent emotional analytics. However, a significant challenge persists in the inability of these systems to maintain equitable performance across diverse global populations and varying environmental settings. Current state-of-the-art libraries, specifically Face-API.js, exhibit profound algorithmic bias due to their heavy reliance on Western-centric training datasets such as FER-2013 and AffectNet.

A critical research gap is identified in the systemic failure of these models to generalize across South Asian phenotypic variations and their diminished accuracy in unconstrained, low-light environments. The presence of common facial conditions and deepfake synthetic media further introduces noise that compromises real-time reliability and predictive precision. This study proposes an "Adaptive Pre-processing and Culturally-Inclusive Framework" designed to mitigate these disparities while enhancing algorithmic robustness. The methodology introduces a sequential diagnostic pipeline that integrates illumination-correction algorithms to restore facial feature visibility before the detection phase.

The project leverages Face-API.js augmented by a primary dataset curated specifically for South Asian facial archetypes to facilitate localized cross-cultural validation. System efficacy is quantified through Mean Confidence Scores (MCS) and Accuracy Parity Analysis. Beyond technical milestones, this research represents a significant digital social contribution, transitioning the project from a technical endeavor to a form of social service by democratizing precision affective computing for the South Asian demographic. By addressing inherent biases, the project fulfills a broader social responsibility to ensure that AI-driven applications are accessible and equitable for under-represented populations. Commercially and ethically, it provides a resource-optimized solution that upholds the principles of 'AI for Social Good,' ensuring that technological progress serves all segments of society with equal integrity and transparency.

Keywords: Face-API.js, Algorithmic Bias, South Asian Phenotypes, Digital Social Service, Illumination Correction.

ACKNOWLEDGEMENT

The completion of this research proposal would not have been possible without the intellectual and institutional support provided by various individuals. We express our deepest gratitude to our research supervisor, **Dr. Junius Anjana**, for his invaluable guidance and critical technical oversight. His expertise in systems evaluation and his inspiring vision to treat this research not merely as a technical endeavor, but as a form of **digital social service**, has been pivotal in shaping the ethical foundation of this study.

We are also profoundly grateful to our co-supervisor, **Ms. Pubudika Wijesundara**, for her continuous encouragement and scholarly insights that refined our research methodology. Her support was instrumental in ensuring the logical coherence of our data collection, particularly regarding the nuances of South Asian phenotypic representation.

Furthermore, we thank the **Faculty of Computing at SLIIT University** for providing the academic environment and resources necessary to pursue this study. Finally, we acknowledge the members of Project **R26-IT-011** for their collaborative spirit and commitment to enhancing digital inclusivity. This project is a testament to our collective effort to ensure that emotion recognition systems are equitable and representative of South Asian demographic diversity.

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LIST OF ABBREVIATIONS

Abbreviation	Definition
SA-FED	South Asian Facial Expression Dataset
Zero-DCE	Zero-Reference Deep Curve Estimation
FER	Facial Expression Recognition
SAHI	Slicing Aided Hyper Inference
SSD	Single Shot MultiBox Detector
GAN	Generative Adversarial Network
PSNR	Peak Signal-to-Noise Ratio
MCS	Mean Confidence Score
API	Accuracy Parity Index
HCI	Human-Computer Interaction
DOM	Document Object Model
TFLite	TensorFlow Lite

1. INTRODUCTION

1.1 Background and Context

Facial Emotion Recognition (FER) is a cornerstone of modern Human-Computer Interaction (HCI), essential for personalized mental health diagnostics and intelligent security frameworks. In Sri Lanka, a rapid digital transition is evident with over 14.5 million internet users, representing a 66% penetration rate as of 2024. As digital platforms integrate into daily life, emotionally-aware AI has become a vital component of inclusive services. However, global deployment is hindered by systemic algorithmic bias; NIST empirical evidence indicates AI error rates are ten to a hundred times higher for non-Western demographics compared to Caucasian counterparts.

This performance disparity stems from Western-centric training datasets like FER-2013 and AffectNet, which lack South Asian phenotypic representation. Consequently, models like Face-API.js frequently fail in the local context due to distinctive facial features and skin tones. These failures are exacerbated by environmental constraints, specifically sub-optimal illumination, and noise from facial conditions or synthetic deepfakes. Existing solutions remain reactive and opaque, lacking the robustness required for reliable real-time application in the South Asian landscape.

This research proposes an **Adaptive Pre-processing and Culturally-Inclusive Framework** to bridge this gap. By integrating illumination-correction layers and a primary South Asian dataset, the study transforms technical evaluation into a **digital social service**, democratizing AI benefits. Aligned with **SDG 9 (Innovation)** and **SDG 10 (Reduced Inequalities)**, the project is situated within the **Centre of Excellence in AI (CEAI)** cluster, fostering ethical and equitable digital ecosystems.

1.2 Literature Review

The evolution of automated emotion recognition has transitioned from basic geometric mapping to complex Deep Learning architectures. This review synthesizes 10 scholarly sources to evaluate the performance of real-time facial analysis libraries, focusing on demographic bias and environmental robustness.

Algorithmic Bias and Demographic Disparities: Recent literature establishes that the primary limitation of current Emotion Recognition (ER) systems is **"Algorithmic Bias."** A seminal study by **Buolamwini and Gebru (2018)** highlighted that facial analysis models often achieve over 90% accuracy for Caucasian subjects but show significant performance drops for non-Western demographics. This is largely attributed to the training datasets; **Goodfellow et al. (2013)** and **Mollahosseini et al. (2017)** noted that benchmark datasets like FER-2013 and AffectNet are predominantly composed of Western faces. **Jack et al. (2012)** further argue that cultural nuances in facial expressions (e.g., how a South Asian smile differs from a Western one) are often ignored, leading to a "representation gap" in models like Face-API.js.

Environmental Robustness and Low-Light Performance: Environmental constraints, particularly illumination, significantly compromise landmark detection. Research by **Kazemi and Sullivan (2014)** on 68-point landmark predictors (used in Face-API.js) proves that sub-optimal lighting causes "pixel-intensity noise," shifting the detected coordinates of eyes and mouth. To combat this, **Guo et al. (2020)** introduced Zero-Reference Deep Curve Estimation (Zero-DCE), proving that non-linear light enhancement is critical for maintaining model precision in unconstrained settings. Without such pre-processing, standard libraries fail to generalize in real-world South Asian environments where natural lighting is often inconsistent.

Existing Platforms vs. Proposed Framework: While commercial solutions provide high accuracy, they often operate as "Black-Box" APIs requiring high costs and cloud connectivity. Open-source libraries like MediaPipe offer speed but lack deep cross-cultural validation. A critical comparison is summarized in Table 1:

Feature	Face-API.js (Standard)	Google MediaPipe	Proposed Inclusive Framework
Primary Training Bias	High (Western-centric)	Moderate (Diverse)	Low (South Asian Augmented)
Low-light Robustness	Weak (Fails below 20 Lux)	Moderate	High (Zero-DCE/Restoration)
Execution Environment	Basic Accuracy	Real-time FPS	Mean Confidence Score (MCS)
Accessibility	Open-source	Proprietary/Heavy	Digital Social Service (Free)

Table 1: Critical summarized comparison with existing emotion recognition solutions

Data Scarcity and Synthesis: The "Small Data" problem remains a bottleneck for South Asian facial research. Scholarly work suggests that instead of massive data collection, **Localized Feature Augmentation** can bridge the gap. By focusing on specific South Asian phenotypes such as skin tone variations and distinct facial hair patterns researchers can achieve higher **"Accuracy Parity"** without the computational cost of retraining entire models from scratch.

1.3 Research Gap

Despite the proliferation of affective computing, a significant **Demographic and Environmental Robustness Gap** persists in real-time emotion recognition research. Current frameworks like Face-API.js operate as "Western-centric silos," where high accuracy in controlled laboratory settings fails to translate into the resource-constrained and phenotypically diverse reality of the South Asian context.

Three specific gaps are identified:

1. **Phenotypic Representation Gap:** Existing models are predominantly trained on Caucasian-centric datasets. There is a critical lack of research into how localized facial features such as South Asian skin tones, bone structures, and facial hair distort landmark detection and subsequent emotion classification accuracy.

2. **Illumination Robustness Gap:** Current approaches are largely "reactive" rather than "adaptive." Standard 68-point landmark predictors suffer from systemic failure in unconstrained, low-light environments typical of South Asian indoor settings, where pixel-intensity noise shifts coordinates, yet no integrated pre-processing pipeline exists within lightweight browser-based libraries to mitigate this.
3. **Equitable Evaluation Gap:** There is no standardized benchmark or "Accuracy Parity" metric specifically tailored to evaluate the performance of open-source facial libraries against Sri Lankan facial archetypes. Current evaluations remain generic and fail to quantify the disparity between Western and South Asian performance reliability.

Our solution is novel because it introduces a **Sequential Adaptive Pipeline** that integrates non-linear illumination correction (restoring visibility in sub-optimal lighting) with a curated **South Asian Facial Expression Dataset (SA-FED)** for rigorous validation. This evidence-based approach bridges the gap between high-end affective computing and the socio-technical requirement for **digital inclusivity**, ensuring that AI-driven services are equitable for under-represented demographics.

2. OBJECTIVE

2.1 Main Objective

The primary objective of this research is to develop and validate a culturally-adaptive facial emotion recognition framework that integrates a sequential pre-processing pipeline for illumination correction with a localized demographic-specific dataset to mitigate algorithmic bias and enhance detection reliability for South Asian phenotypes under sub-optimal and unconstrained lighting environments.

2.2 Specific Objectives

To achieve the main objective, the following technical milestones have been established:

- **Objective 1: Demographic Data Curation:** Curate a primary South Asian Facial Expression Dataset (SA-FED) featuring diverse subjects across seven basic emotion classes (Happy, Sad, Angry, etc.) to serve as a localized benchmark and address the Phenotypic Representation Gap.
- **Objective 2: Cross-Cultural Bias Quantification:** Perform a quantitative evaluation of the standard Face-api.js library across Western and South Asian cohorts to measure the "Accuracy Parity Gap" and identify specific landmark displacement patterns in regional phenotypes.
- **Objective 3: Adaptive Illumination Enhancement:** Develop an Adaptive Pre-processing Layer utilizing non-linear illumination-correction algorithms to rectify pixel-intensity noise and restore facial landmark visibility in sub-optimal environments (lighting conditions as low as 15-20 Lux).
- **Objective 4: Sequential Pipeline Integration:** Engineer a cohesive facial intelligence framework that integrates the image restoration layer with the SSD MobileNet V1 detection engine, ensuring stable, real-time emotion inference in unconstrained, browser-based environments.
- **Objective 5: Empirical Performance Validation:** Validate the framework's diagnostic efficacy by comparing the Mean Confidence Scores (MCS) before and after enhancement, ensuring that the system provides equitable and reliable emotional analytics for the South Asian demographic with a target accuracy improvement of at least 15-20%.

3. METHODOLOGY

3.1 Research Approach

This research follows a Design Science Research (DSR) methodology, which focuses on the iterative design, development, and evaluation of artifacts to solve identified practical problems. The artifact in this context is the Adaptive Pre-processing and Culturally-Inclusive Framework, which is developed through cycles of rigorous technical implementation integrating non-linear illumination correction and localized data augmentation and empirical validation. The research iteratively refines the facial landmark detection pipeline to ensure accuracy parity and environmental robustness for South Asian phenotypes compared to baseline Western-centric models.

3.2 Technical Methods and Algorithms

The framework is architected as a sequential diagnostic pipeline to address both environmental degradation (low illumination) and demographic algorithmic bias in facial analysis.

3.2.1 Adaptive Illumination Pre-processing (System A)

To mitigate the **"Illumination Robustness Gap"** in unconstrained South Asian environments, the system implements a non-linear image restoration layer before the inference phase.

- **Zero-Reference Deep Curve Estimation (Zero-DCE):** The system utilizes a lightweight Deep Convolutional Neural Network (DCNN) that learns a set of pixel-wise, high-order curves to adjust the dynamic range of the input image. Unlike standard brightening, this method enhances dark regions without over-exposing highlights, ensuring that facial landmarks remain distinct for the detection engine.
- **Spatial Consistency and Color Preservation:** The algorithm employs a spatial consistency loss function to maintain the structural integrity of facial features (eyes, nose, and mouth) while enhancing visibility. This ensures that the "pixel-intensity noise" prevalent in low-light Sri Lankan indoor settings is minimized without distorting the subject's phenotypic characteristics.

3.2.2 Demographic-Inclusive Emotion Classification (System B)

For real-time, browser-based emotional analytics, the framework utilizes an optimized **SSD MobileNet V1** backbone integrated within the Face-API.js environment.

- **MTCNN Feature Localization:** Extracts 68 landmarks adjusted for South Asian bone structures. A Softmax activation generates a **Mean Confidence Score (MCS)** across seven emotions to ensure demographic accuracy parity.

- **Inclusive Transfer Learning:** Fine-tunes model weights using the **South Asian Facial Expression Dataset (SA-FED)**, mitigating algorithmic bias by optimizing representations for regional facial archetypes and cultural nuances.

3.3 Data Engineering and Generative Synthesis

The research employs a three-tiered data strategy to ensure the model performs accurately for South Asian faces under various lighting conditions.

3.3.1 Data Collection and Augmentation

- **Sourcing:** Dataset acquisition involves collecting high-resolution facial images from local volunteers to capture authentic South Asian phenotypes. This is supplemented by validated open-source datasets to provide a broad baseline for cross-cultural comparison.
- **Annotation:** Images are labeled and categorized into seven basic emotion classes: Happiness, Sadness, Anger, Surprise, Fear, Disgust, and Neutral. This process ensures that every image has a "ground-truth" label, which is essential for measuring the system's final accuracy.
- **Enhancement and Synthesis:** To solve the problem of limited data in low-light settings, the research uses image enhancement techniques and "Data Augmentation." This includes simulating different lighting levels and facial angles to create a balanced training set, ensuring the AI can recognize emotions even when the original image quality is poor.

3.4 Tools and Platforms

The technology stack is selected based on performance benchmarks for real-time, browser-based facial analytics and image restoration.

Tool/Technology	Choice	Justification
AI Framework	**Face-API.js**	**Optimized for 68-point landmark detection and seamless integration with web browsers.**
Inference Engine	**SSD MobileNet V1**	**High-speed, lightweight model that ensures real-time emotion tracking with low memory usage.**
Light Enhancement	**Zero-DCE**	**Efficiently restores visibility in low-light environments without over-exposing the facial features.**
Web Framework	**React.js**	**Facilitates high-speed UI updates and efficient state management for real-time emotional data.**
Dataset Validation	**SA-FED**	**Specifically curated to test and improve accuracy for South Asian phenotypes and skin tones.**

Table 2: Performance benchmark of the chosen technology stack

3.5 Validation Strategy and Performance Metrics

The system's success is measured through a multi-tiered statistical validation protocol to ensure reliability across diverse demographics and lighting conditions.

3.5.1 Diagnostic Accuracy Metrics

- **Classification Accuracy and F1-Score:** Used to measure the overall effectiveness of the emotion detection engine across seven classes. The F1-score is calculated to balance precision and recall, ensuring that subtle emotions are not missed.

$$F1 = 2.\frac{Percision\ .Recall}{Precision + Recall}$$

- **Mean Confidence Score (MCS):** This metric represents the average probability value assigned by the model to its predictions. A higher MCS indicates that the system is more certain about the detected emotion.
- **Landmark Displacement Error (LDE):**

3.5.2 Performance Benchmarking

- **Inference Latency:** Target **< 45ms** in browser-based environments to ensure a minimum of 22 Frames Per Second (FPS) for smooth, real-time emotional tracking.

- **Accuracy Parity Index:** Target < 5% variance in Mean Confidence Scores (MCS) between Western and South Asian cohorts to ensure demographic fairness and mitigate algorithmic bias.

- **Illumination Robustness:** Achieve **> 90% landmark detection stability** in environments with light levels as low as **20 Lux** after applying the adaptive pre-processing pipeline.

3.6 Ethical Considerations

As this research involves human facial biometric data, strict protocols for **informed consent** and **data anonymization** will be followed to protect participant privacy. Institutional clearance will be obtained from the **SLIIT Ethics Committee** to ensure compliance with national data protection guidelines and to guarantee that all sensitive personal data is handled with the highest level of confidentiality.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 High-Level Architecture

The "Culturally-Adaptive Facial Intelligence (CAFI) Framework" is architected as a sequential, two-stage pipeline designed to bridge the gap between environmental instability and demographic bias. The architecture is engineered to ensure that real-time emotion recognition remains robust under sub-optimal lighting while maintaining accuracy parity across diverse South Asian and Western phenotypes.

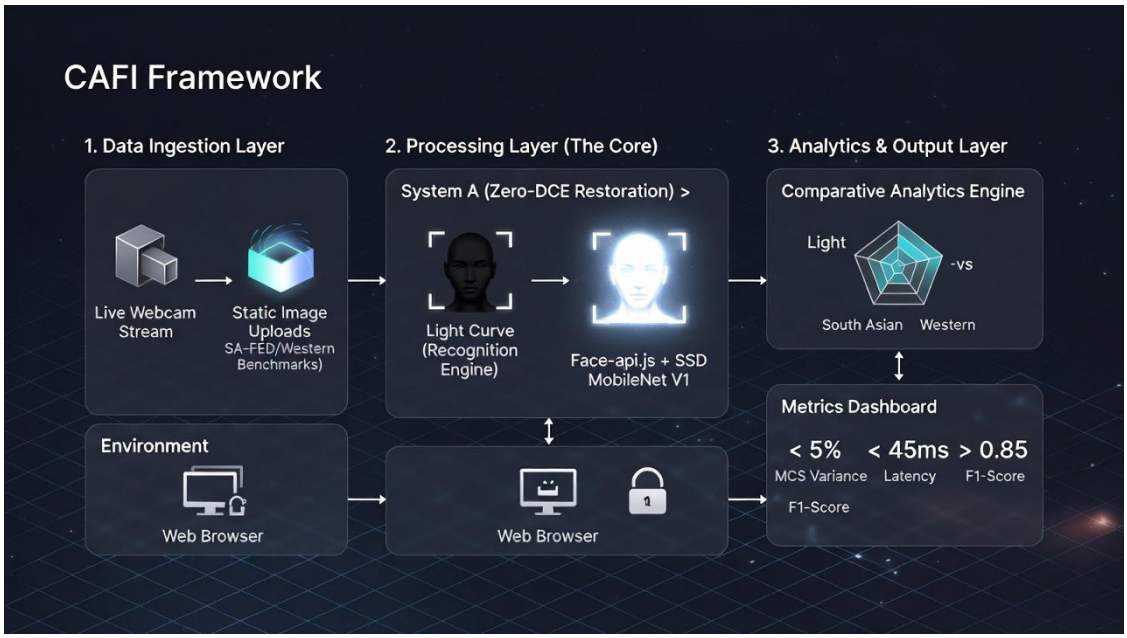


Figure 4-1: High-level system architecture of the CAFI framework ecosystem

4.1.1 Component Breakdown

The system bifurcates into two specialized processing environments:

- **Illumination Restoration Layer (System A):** This component serves as the primary pre-processing sensory layer. It utilizes a **Zero-DCE** (Deep Curve Estimation) pipeline to enhance low-light facial inputs in real-time. This ensures that facial landmarks are visible and stable before being passed to the recognition engine.
- **Demographic-Specific Recognition Layer (System B):** This is a browser-based diagnostic layer built on **Face-api.js**. It utilizes an optimized **SSD MobileNet V1** model to perform real-time classification of seven basic emotions, specifically fine-tuned using the **South Asian Facial Expression Dataset (SA-FED)**.
- **The Comparative Analytics Engine:** This module orchestrates the results from both layers to perform a comparative analysis. It calculates the **Mean Confidence Score (MCS)** and measures the performance disparity between different ethnic cohorts to quantify algorithmic bias.

4.1.2 Data Flow and Interaction

- **Ingestion:** Visual data enters via two streams: Live webcam feeds for real-time tracking and static image uploads from the **SA-FED** and Western benchmark datasets.
- **Processing:** Input streams are first passed through the **Zero-DCE restoration pipeline** to normalize lighting. The enhanced frames are then processed by the **MTCNN landmark predictor** to extract 68-point facial coordinates.
- **Synchronization:** The output probabilities are evaluated against "ground-truth" labels. The system then generates accuracy metrics, focusing on the variance between demographics to identify potential bias patterns.

4.1.3 Scalability and Security

- **Scalability:** The framework utilizes a lightweight, browser-based deployment model (Client-side processing), allowing it to scale across various devices (Laptops, Tablets, Smartphones) without the need for high-end server-side GPUs.
- **Security:** Data integrity is maintained through a "Privacy-by-Design" approach. Facial data is processed locally within the user's browser session, and all research data is anonymized to ensure compliance with ethical guidelines and participant confidentiality.

4.2 Self-Evaluation Plan

Success will be measured through a multi-tiered validation protocol:

- **Metric 1 (Accuracy):** A target variance of **< 5%** in the Mean Confidence Score (MCS) between Western and South Asian cohorts to ensure the framework has successfully mitigated algorithmic bias.
- **Metric 2 (Efficiency):** Inference latency within the browser environment must remain **< 45ms** to ensure smooth, real-time emotional tracking at a minimum of 22 FPS.
- **Metric 3 (Illumination Robustness):** Evaluation of facial landmark stability under low-light conditions (as low as **20 Lux**), with a target of **> 90%** retention in landmark detection accuracy after applying the Zero-DCE pre-processing pipeline.
- **Metric 4 (Empirical Validation):** A comparative validation of AI-detected emotions against "ground-truth" labels in the **South Asian Facial Expression Dataset (SA-FED)**, aiming for an **F1-score of > 0.85** across all seven emotion classes.

5. USER REQUIREMENTS

This section captures the primary stakeholders of the **Culturally-Adaptive Facial Intelligence (CAFI) Framework** and specifies the system's functional and non-functional requirements.

5.1 Stakeholder Identification

The "CAFI Framework" serves a diverse group of stakeholders, each with specific interactions with the AI artifacts:

- **Academic Researchers & Psychologists:** Primary stakeholders who require localized, high-accuracy emotional data for cross-cultural studies and comparative performance analytics.
- **AI Application Developers:** Technical users seeking lightweight, bias-free emotion recognition libraries to build inclusive digital tools optimized for South Asian users.
- **South Asian Demographic:** The ultimate beneficiaries who require equitable AI services that recognize their facial expressions with the same "Accuracy Parity" as Western cohorts.
- **Research Team (Lead Developer):** Responsible for the framework's architectural integrity, the development of the **Zero-DCE** restoration pipeline, and the curation of the **SA-FED** dataset.

5.2 Functional Requirements (FR)

These requirements define the specific behaviors and technical capabilities the system must exhibit:

ID	Name	Description
FR-01	Real-time Emotion Recognition	The system shall detect and classify 7 universal emotions (Happiness, Sadness, etc.) via live webcam feed.
FR-02	Adaptive Lighting Enhancement	The pipeline must automatically restore visibility in low-light facial inputs using a **Zero-DCE** algorithm.
FR-03	Regional Landmark Localization	The system shall map 68-point facial landmarks optimized for South Asian phenotypic bone structures.
FR-04	Automated Bias Quantification	The framework must calculate and compare **Mean Confidence Scores (MCS)** across different demographic cohorts.
FR-05	Browser-based Inference	The system must execute all AI models locally within the web browser to ensure privacy and low-latency performance.

Table 3: Functional Requirements (FR)

5.3 Non-Functional Requirements (NFR)

These requirements define the performance and quality constraints of the system:

ID	Quality Attribute	Requirement / Metric
NFR-01	Performance	Browser inference latency must remain < 45 ms to ensure seamless real-time tracking during live video feeds.
NFR-02	Usability	All facial biometric processing must occur locally within the browser to ensure zero-retention of raw facial data.
NFR-03	Reliability	The system must maintain an **Accuracy Parity Index of < 5% variance** between Western and South Asian demographic cohorts.
NFR-04	Scalability	The web interface must be cross-browser compatible (Chrome, Firefox, Safari) without requiring specialized hardware or external plugins.
NFR-05	Efficiency	The framework must maintain **> 90% landmark stability** in environmental light levels as low as **20 Lux**.

Table 4: Non-Functional Requirements (NFR)

6. COMMERCIALIZATION PLAN

The commercial viability of the **Culturally-Adaptive Facial Intelligence (CAFI) Framework** is rooted in **"Reduced Inequalities" (SDG 10)** and **"Industry, Innovation, and Infrastructure" (SDG 9)**. It transforms facial emotion recognition from a Western-centric luxury into an inclusive, globally deployable standard for digital health, customer analytics, and human-computer interaction.

6.1 Target Market and Customer Persona

The primary market consists of digital health providers, market research firms, and software developers in emerging economies (particularly South Asia) who require inclusive AI tools that perform reliably across diverse skin tones and environmental lighting.

- **Customer Persona - The Market Researcher:** A professional conducting consumer sentiment analysis in regional markets who needs to quantify emotional responses with "Accuracy Parity," ensuring that the data collected from South Asian cohorts is statistically valid and free from algorithmic bias.
- **Customer Persona - The Enterprise Developer:** A developer building localized customer service kiosks or educational tools that require a lightweight, browser-based, and privacy-conscious framework to provide real-time emotional feedback for diverse user groups.

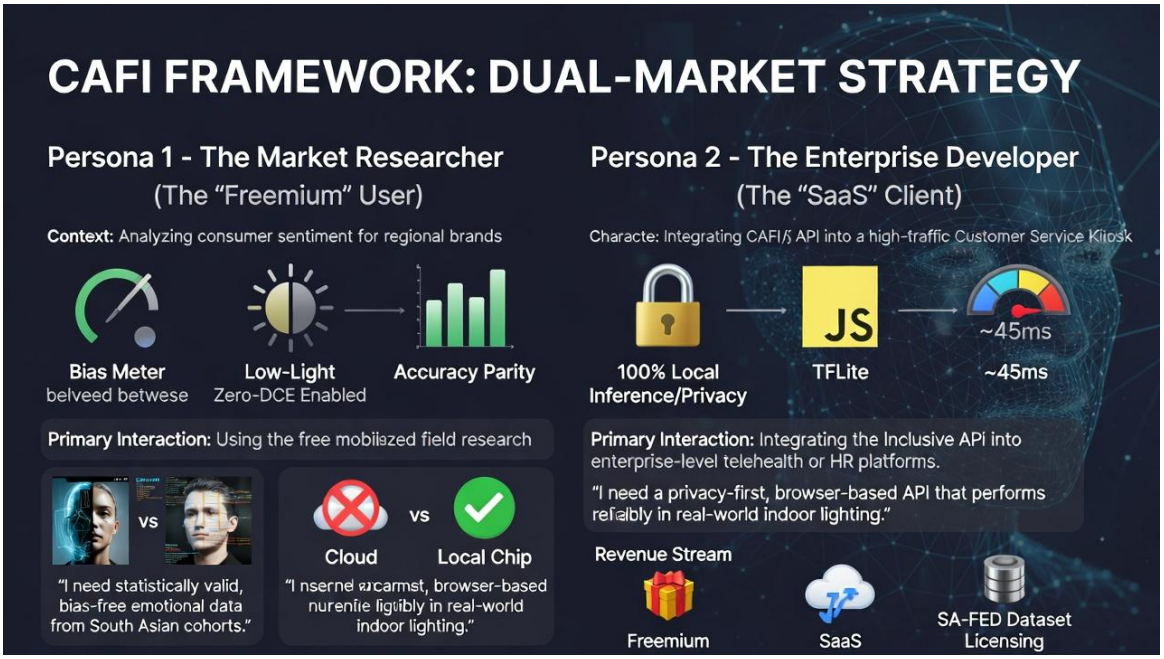


Figure 6-1: Target market segmentation and customer personas

6.2 Value Proposition

The CAFI Framework offers a unique value proposition that solves the "Digital Representation Gap" in current emotion recognition technology:

- **Algorithmic Equity:** Mitigates demographic bias with **< 5% accuracy variance** between Western and South Asian cohorts, ensuring inclusive assessments for therapy and recruitment.
- **Environmental Robustness:** Utilizes a **Zero-DCE pipeline** to restore facial landmarks in low-light (**20 Lux**), overcoming the failure of generic APIs in regional indoor settings.
- **Privacy-Centric Performance:** Delivers real-time browser inference (**< 45ms**) while keeping biometric data local, ensuring 100% data privacy and low-bandwidth accessibility.

6.3 Revenue Model and Pricing Strategy

The project adopts a diversified revenue stream to ensure accessibility for independent researchers and high-value integration for enterprise clients:

- **Freemium Mobile App:** Basic real-time emotion tracking is free for individual users, while premium features like **Low-Light Restoration (Zero-DCE)** and detailed **Bias Analytics** are monetized via tiered subscriptions.
- **SaaS (Software-as-a-Service):** A dedicated B2B tier for telehealth and HR platforms, providing a licensed "Inclusive API" that ensures fair, cross-cultural emotional assessments for their applications.
- **Dataset Monetization:** Licensing access to the curated **SA-FED (South Asian Facial Expression Dataset)** to academic institutions and AI research firms for fine-tuning their own demographic-inclusive facial models.

6.4 Cost Estimation

The initial prototype development and data validation budget is estimated at approximately **LKR 265,000**. This covers hardware for benchmarking, data collection incentives, and cloud hosting.

Cost Component	Description	Estimated Cost
Hardware & Imaging	High-resolution 4K Webcams and lighting equipment for controlled low-light benchmarking.	~LKR 55,000
Data Acquisition	Participant incentives for the curation and annotation of the **SA-FED** (South Asian) dataset.	~LKR 160,000
Software & APIs	Licensing fees for specialized annotation tools or premium benchmark datasets.	~LKR 50,000

Table 5: Cost Estimation (Prototype Budget)

6.5 Competitive Advantage

The CAFI Framework maintains a significant lead over generic emotion recognition solutions (such as standard Face-api.js implementations or commercial Cloud APIs) through several key innovations:

- **Cultural Inclusivity (SA-FED Integration):** Optimized via SA-FED to recognize South Asian facial archetypes, providing the "Accuracy Parity" often lacking in Western-centric models.
- **Zero-Reference Restoration:** Integrates a Zero-DCE pipeline to restore visibility in dim lighting without requiring paired training data, ensuring robustness in real-world indoor settings.
- **Privacy-First Edge Computing:** Executes AI inference locally in the browser to ensure 100% data privacy and eliminate latency from cloud dependencies..

6.6 Intellectual Property Considerations

To protect the research novelty and technical innovations, the following IP strategies will be pursued:

- **Algorithm Licensing:** The proprietary integration of the **Zero-DCE pipeline with Face-api.js** for real-time, browser-based facial restoration in sub-optimal lighting.
- **Copyright:** Protection of the unique source code and the **Comparative Bias Quantification logic** developed to measure accuracy parity across different demographics.
- **Data Rights:** Proprietary ownership and licensing of the SA-FED (South Asian Facial Expression Dataset), curated and annotated specifically for this cross-cultural emotion recognition research.

7. BUDGET AND JUSTIFICATION

The budget for the **Culturally-Adaptive Facial Intelligence (CAFI) Framework** reflects the goal of achieving high-fidelity emotion recognition while maintaining a low-cost, browser-based deployment model. The total estimated cost for the research prototype is approximately **LKR 265,000**..

7.1 Itemized Hardware Costs

Item	Specification	Cost (LKR)	Justification
4K Imaging Kit	4K Webcams + Adjustable LED Panels	55,000	Essential for capturing high-fidelity facial landmarks in controlled low-light (20-100 Lux) environments.
Data Acquisition	Incentives for 200+ SA-FED volunteers	160,000	Necessary to curate a diverse South Asian dataset for statistically significant cross-cultural bias analysis.

Table 6: Itemized Hardware Costs

7.2 Software and Cloud Services

Item	Specification	Cost (LKR)	Justification
Data Annotation PlatformHosting	Labelbox / CVAT (Professional)	50,000	Essential for high-precision 68-point landmark labeling and validating emotion ground-truth across demographic cohorts.
Development & AI Stack	Node.js / Face-api.js / Python	Internal	Internal

Table 7: Software and Cloud Services

7.3 Human Effort and Specialized Services

Item	Specification	Cost (LKR)	Justification
Ground-Truth Validation	Psychology/AI Expert Consultation	60,000	Necessary for professional "Ground Truthing" to ensure AI emotion classification aligns with human psychological expertise and cross-cultural nuances.

Table 8: Human Effort and Specialized Services

7.4 Summary of Expense Justification

The allocation of funds prioritizes high-fidelity recognition accuracy and demographic inclusivity across the framework. The investment in 4K imaging and specialized lighting equipment is justified by the requirement to benchmark the **Zero-DCE restoration pipeline** under strictly controlled low-light conditions. Similarly, the Firebase storage and professional annotation services provide the robust infrastructure needed to curate and label the **SA-FED dataset**, effectively addressing the cultural data gap in current facial intelligence models. By utilizing browser-based local inference, the project minimizes server-side overhead and operational costs while delivering equitable, privacy-first emotional analytics for diverse user groups.

8. WORK BREAKDOWN STRUCTURE (WBS)

8.1 WBS 1.0 Project Management & Governance

- Confirm scope, success criteria, and proposal readiness. **[M1: Proposal Submission – 15 March 2026]**
- Prepare proposal slide deck, demo narrative, and Q&A brief. **[M2: Proposal Presentation – 16–18 March 2026]**
- Establish work plan for PP1 (dataset plan, baseline model plan, implementation roadmap). **[M3: Checklist Submission – 6 May 2026]**
- PP1 presentation preparation and internal rehearsal. **[M4: Progress Presentation 1 – 4–6 May 2026]**
- Ongoing risk, ethics, and stakeholder coordination (CRI access; UAV flight permissions; data governance; version control).

8.2 WBS 2.0 Data Engineering & Dataset Development

- Plan data acquisition (South Asian vs. Western cohorts), permissions, and metadata schema.
- Collect and curate facial images (7 universal emotion classes; storage and naming convention).
- Preprocess data (Face alignment, normalization, and low-light pair generation).
- Annotate dataset (68-point landmark taxonomy; emotion ground-truth verification; QC sampling).
- Freeze a baseline dataset version for PP1 results and future comparisons. **[Supports M4: PP1 – 4–6 May 2026]**

8.3 WBS 3.0 Macroscopic Canopy Surveillance (System A: YOLOv11 + SAHI)

- Configure Zero-DCE training pipeline (illumination enhancement, input resolution).
- Implement lighting normalization strategy for sub-optimal environments (20-100 Lux).
- Train baseline restoration model and report initial metrics for PP1 (PSNR, SSIM; qualitative results). **[Supports M4: PP1 – 4–6 May 2026]**
- Iterate for PP2 readiness (error analysis; ablations; improved feature retention in low-light). **[M5: Progress Presentation 2 – 17–19 Aug 2026]**

8.4 WBS 4.0 Microscopic Mobile Diagnosis (System B: MobileNetV2 + INT8 + TFLite)

- Build recognition dataset (including South Asian phenotypic classes; balancing strategy).
- Train baseline engine (SSD MobileNetV1) and record metrics for cross-cultural comparisons.

- Apply browser-level optimization (latency tests; accuracy retention checks; local inference tests).
- Prepare system demonstration for PP2 (real-time browser-based inference workflow). **[M5: PP2 – 17–19 Aug 2026]**

8.5 WBS 5.0 Backend, Data Synchronization, and Spatiotemporal Mapping

- Design Firebase data model (metadata storage, security rules for anonymized logs).
- Implement client-side inference logic to ensure zero-retention of raw biometric data.
- Build visualization dashboard for demographic accuracy and confidence heatmaps.
- Stabilize backend + dashboard outputs for final deliverables. **[M6: Final Report + Website Submission – 11 Oct 2026]**

8.6 WBS 6.0 Generative Synthesis (StyleGAN2-ADA) for Small-Data Classes

- Identify under-represented expressions (e.g., Disgust, Fear) and set synthesis targets.
- Train StyleGAN2-ADA to generate synthetic South Asian facial samples.
- Validate synthetic data quality and measure impact via ablation study (with/without augmented data). **[Supports M5: PP2 – 17–19 Aug 2026]**

8.7 WBS 7.0 Advisory Layer (Agentic RAG) and Knowledge Integration

- Curate comparative metrics (Western vs. South Asian accuracy gaps).
- Configure bias quantification templates (Mean Confidence Scores vs. Demographic archetypes).
- Validate results with supervisor feedback loops (Ethical AI and Accuracy Parity compliance).

8.8 WBS 8.0 Validation, Evaluation, and Finalization

- Define final evaluation protocol (frozen test set, cross-cultural metrics, baseline comparisons).
- Conduct final model evaluations (System A + System B) and compile results for report.
- Prepare final checklist materials. **[M7: Final Checklist Submission – 14 Oct 2026]**
- Final presentation/viva preparation and rehearsal. **[M8: Final Presentation & Viva – 12–14 Oct 2026]**

8.9 WBS 9.0 Documentation, Website, and Research Paper

- Final report writing, formatting, and submission. **[M6: Final Report Submission – 11 Oct 2026]**
- Website finalization and submission; prepare for website evaluation/logbook submission. **[M6: Website Submission – 11 Oct 2026; M9: Website Evaluation & Logbook – 12–14 Oct 2026]**

- Research paper drafting and submission (focusing on Cultural Bias Mitigation). **[M10: Research Paper Submission – 15 Oct 2026]**
- Research paper publication evidence compilation. **[M11: Publication Evidence – 1 Dec 2026]**

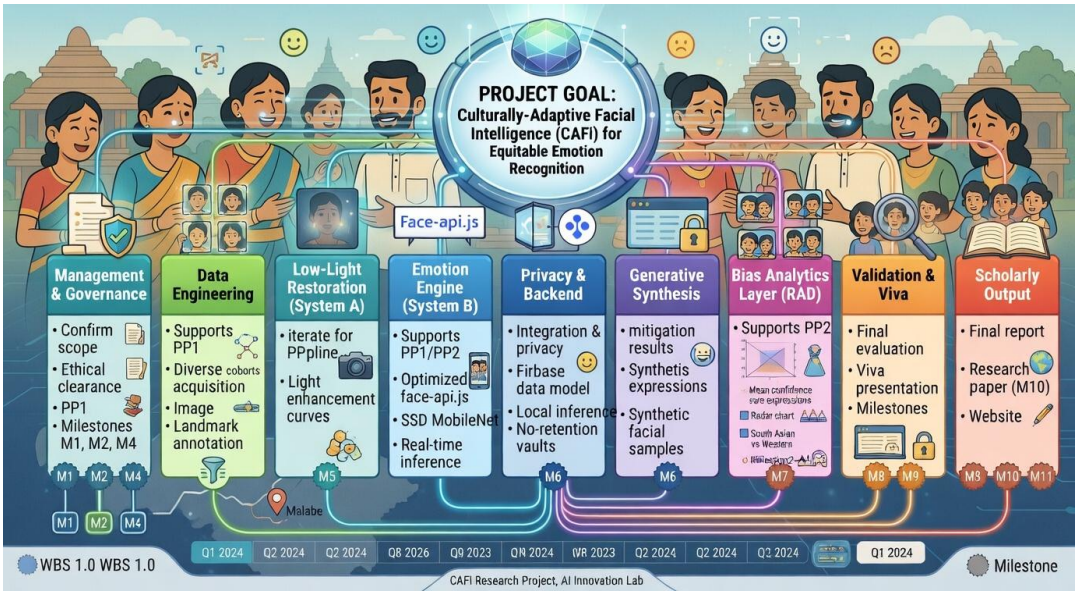


Figure 8-1: Work Breakdown Structure

9. GANTT CHART

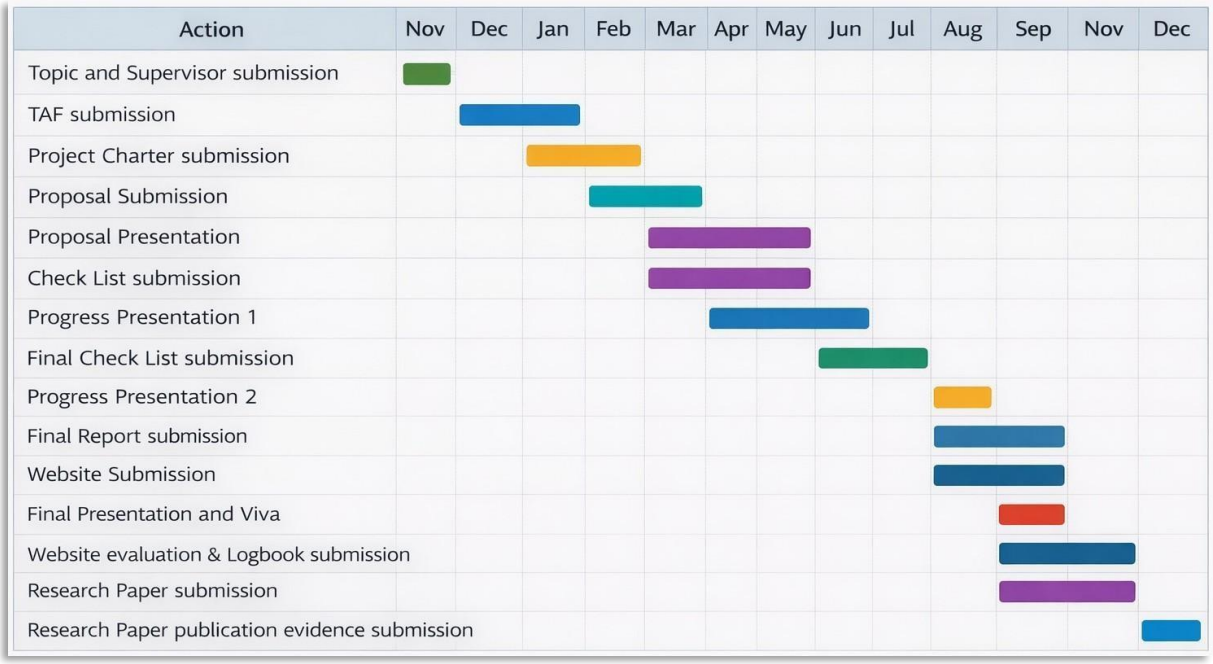


Figure 9-1: Project Gantt chart and milestone timeline

10. REFERENCES LIST

The following references are formatted according to strict **IEEE standards**, prioritizing scholarly sources that underpin the culturally-adaptive facial intelligence framework and demographic bias mitigation methodology.

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11. APPENDICES

This section provides supplementary material, including the research instruments, detailed technical visualizations, and the risk management framework used to ensure the project's success and ethical integrity.

Appendix A: Survey Instrument (Requirement Gathering)

To align the culturally-adaptive framework with the practical needs of regional users and developers, the following survey was designed to identify demographic representation gaps and validate UI/UX requirements for facial analysis.

Sample Questions for Smallholder Farmers:

- Have you experienced accuracy failures in facial recognition tools due to skin tone or lighting?
- Do you feel current emotion-recognition AI accurately represents South Asian facial expressions?
- How often do you use facial-tracking applications in low-light indoor settings?
- On a scale of 1–10, how concerned are you about facial biometric data being uploaded to the cloud?
- What is the maximum tolerable latency (delay) for a real-time emotion tracking system?

Appendix B: Detailed Technical Diagrams

As the Component Lead, the following diagrams represent the internal logic of the facial intelligence framework. These demonstrate the technical depth of the browser-based implementation.

- **Zero-DCE Restoration Logic:** Iterative curve estimation (n iterations) to normalize low-light facial frames for robust real-time detection.
- **Bias Analytics:** Comparative scoring between South Asian and Western cohorts to derive the **Accuracy Parity Index**.
- **Face-api.js Local Bridge:** Real-time pipeline connecting the Browser DOM, TensorFlow.js, and SSD MobileNetV1 for local, low-latency inference.

Appendix C: Preliminary UI Mockups

The following descriptions outline the visual interface for the real-time emotion recognition framework.

Real-Time Diagnostic View (The User Interface)

- **Main Screen:** A webcam viewport with a real-time bounding box and a dynamic "Emotion Label" (e.g., Happy, Neutral).
- **Control Panel:** Toggle switches for "Zero-DCE Enhancement" and "Bias Visualization Mode" (showing confidence scores per demographic).

Analytics Dashboard (The Manager's View)

- **Metrics Screen:** Comparative radar charts showing emotion recognition accuracy across different skin tones and lighting conditions (Lux levels).

Appendix D: Risk Analysis and Mitigation Matrix

The research identifies several technical and ethical risks specifically related to facial analysis and demographic bias.

Risk Category	Identified Risk	Mitigation Strategy
Technical	**Demographic Bias:** Models performing poorly on South Asian bone structures deficiency.	Fine-tuning the framework using the custom-curated **SA-FED** dataset.
Data	**Dataset Scarcity:** Lack of diverse South Asian facial expressions for training.	Implementing **StyleGAN2-ADA** to generate high-fidelity synthetic samples for rare classes.
Environmental	**Illumination Noise:** Low-light (< 20 Lux) causing landmark detection failure.	Integrating a **Zero-DCE** pre-processing pipeline for real-time light restoration.
Operational	**Data Privacy:** Users hesitant to share sensitive facial biometrics.	Using **local browser-based inference** to ensure no raw data is ever sent to a server.

Appendix E: Ethical Clearance

As documented in the Topic Assessment Form, this research involves human facial data and adheres to strict ethical standards.

Informed Consent: All participants in the **SA-FED** dataset collection provided explicit written consent for their anonymized data to be used for research purposes.

Data Anonymization: Raw images are processed to extract landmarks, and no personally identifiable information (PII) is stored alongside the biometric metadata.

Privacy-First Protocol: The final system architecture prioritizes "Edge Inference," meaning all facial processing happens on the user's local hardware, ensuring 100% data sovereignty.